

Project Initialization and Planning Phase

Date	29 June 2026
Team ID	SWTID-2026-4993
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements :

Farmers often face difficulties in selecting the most suitable crop due to changing soil conditions, weather variations, and the lack of accurate decision-making tools. Traditional farming methods and guesswork can result in poor crop yield, financial losses, and inefficient use of natural resources. To address this problem, the OptiCrop system uses machine learning to analyze soil nutrients (Nitrogen, Phosphorus, Potassium) and environmental factors such as temperature, humidity, pH, and rainfall, providing farmers with accurate crop recommendations. This helps improve agricultural productivity, optimize resource utilization, and support sustainable farming practices.

Agriculture plays a vital role in ensuring food security and economic growth. However, many farmers still rely on traditional knowledge, personal experience, or guesswork when selecting crops for cultivation. Due to changing weather conditions, variations in soil fertility, and limited access to scientific recommendations, farmers often choose crops that are not suitable for their land. This results in low crop yields, inefficient use of fertilizers and water, increased cultivation costs, and financial losses. In addition, improper crop selection can degrade soil quality and negatively impact long-term agricultural sustainability.

To overcome these challenges, there is a need for an intelligent, data-driven system that can analyze important agricultural parameters such as soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH, and rainfall. The **OptiCrop** system addresses this problem by using machine learning algorithms to recommend the most suitable crop for cultivation based on these parameters. This helps farmers make informed decisions, improve productivity, optimize resource utilization, reduce farming risks, and promote sustainable agricultural practices.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A small-scale farmer in rural India	Identify the best crop to grow this season	I don't know which crop suits my soil's N, P, K, pH, and climate	I lack access to agricultural experts and scientific tools	Frustrated, financially stressed, and uncertain
PS-2	An agricultural researcher	Analyze crop-environment relationships for policy planning	Existing tools don't provide data-driven pattern insights	Most systems are generic and not ML-powered	Limited in making evidence-based recommendations
PS-3	A government policymaker	Design sustainable farming strategies for a region	There is no centralized system linking soil data to crop productivity	Agricultural data is scattered and unprocessed	Overwhelmed and unable to plan resource allocation efficiently